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Hospital Patient Records Cleaning and health Metrics Dashboard

Project Domain

[Power BI with Copilot]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

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ABSTRACT

This project has been developed under the AICW Fellowship Program organized by Edunet Foundation in collaboration with Microsoft and LinkedIn. The primary objective of this project is to demonstrate how data analytics and business intelligence tools can be applied in the healthcare domain to improve decision-making and operational efficiency.

Hospitals generate a vast amount of patient-related data on a daily basis, including demographic details, diagnosis records, treatment history, and admission and discharge information. However, this data is often unstructured, inconsistent, and contains errors such as missing values, duplicate entries, and incorrect formatting. These issues reduce the reliability of data and make it difficult for hospital administrators to extract meaningful insights.

This project focuses on cleaning hospital patient records and developing a Health Metrics Dashboard using Microsoft Excel and Power BI. Data preprocessing techniques such as removing duplicates, handling missing values, and standardizing data formats were applied to ensure data accuracy and consistency.

After cleaning, key healthcare performance indicators such as total number of patients, department-wise patient distribution, average length of hospital stay, treatment success rate, and admission vs discharge trends were calculated. These metrics were visualized through an interactive dashboard, enabling easy interpretation of hospital performance.

Additionally, Artificial Intelligence tools such as ChatGPT and Microsoft Copilot were used to assist in data analysis, interpretation, and report generation. The final outcome of the project demonstrates how cleaned and structured data, combined with visualization tools, can support effective decision-making in healthcare management.

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CHAPTER 1: INTRODUCTION

In the modern healthcare system, data plays a crucial role in improving patient care and operational efficiency. Hospitals continuously generate large volumes of data through patient admissions, diagnoses, treatments, and discharge processes. However, raw data in its original form is often difficult to analyze due to inconsistencies and lack of structure.

Data analytics and business intelligence tools have made it possible to transform raw healthcare data into meaningful insights. Tools such as Microsoft Excel and Power BI allow users to clean, process, analyze, and visualize data effectively. These insights can help hospital administrators monitor patient flow, evaluate treatment outcomes, and optimize resource utilization.

This project focuses on cleaning hospital patient records and creating a Health Metrics Dashboard. The aim is to demonstrate how data analytics can be applied in the healthcare sector to improve decision-making and enhance hospital performance.

1.1 Problem Statement

Hospitals generate a large amount of patient data every day, including patient demographics, admission and discharge details, diagnosis records, and treatment outcomes. However, this data is often stored in raw formats such as spreadsheets or hospital management systems, which may contain inconsistencies and errors.

Common issues found in hospital datasets include missing patient details, duplicate records, inconsistent data formats, and incorrect entries. These issues make it difficult for healthcare professionals to analyze the data accurately and derive meaningful insights.

Due to poor data quality, hospital administrators may face challenges in tracking patient flow, evaluating treatment success rates, and managing hospital resources efficiently. This can ultimately affect the quality of healthcare services provided to patients.

Therefore, there is a need for an effective system that can clean and process hospital data and present it in a structured and visual format. This project aims to address these challenges by developing a Hospital Patient Records Cleaning and Health Metrics Dashboard.

1.2 Motivation

The motivation behind this project is to apply data analytics techniques in the healthcare domain, where accurate data plays a critical role in decision-making. Efficient management of hospital data

can significantly improve patient care, reduce operational inefficiencies, and enhance overall healthcare performance.

With the increasing importance of data-driven decision-making, it becomes essential to use tools that can simplify data analysis and visualization. This project was undertaken to understand how tools like Excel, Power BI, and AI-based assistants can be used to analyze real-world datasets.

Additionally, the project aims to bridge the gap between theoretical knowledge and practical application by providing hands-on experience in data cleaning, analysis, and dashboard development.

1.3 Objectives

The primary objective of this project is to develop a system that can clean hospital patient data and present it through an interactive dashboard for better analysis.

The project aims to preprocess raw datasets by removing inconsistencies, handling missing values, and standardizing data formats. It also focuses on analyzing patient data to identify patterns and trends in hospital operations.

Another important objective is to calculate key healthcare metrics such as total number of patients, average hospital stay, department-wise distribution, and treatment success rate. These metrics help in evaluating hospital performance.

The project also aims to design an interactive dashboard using Power BI that visually represents the analyzed data. Furthermore, it includes the use of AI tools to assist in analysis and documentation, improving efficiency and understanding.

1.4 Scope of the Project

This project is limited to the analysis of hospital patient data using Microsoft Excel and Power BI. The dataset used includes fields such as patient ID, age, gender, diagnosis, admission date, discharge date, department, and treatment outcome.

The scope of the project includes data cleaning, descriptive data analysis, and dashboard creation. It does not involve real-time data integration or advanced machine learning techniques.

The project is designed to demonstrate how basic data analytics tools can be used to improve healthcare management and decision-making. It provides a foundation for future enhancements such as predictive analytics and real-time monitoring systems.

CHAPTER 2: LITERATURE SURVEY

Various studies and research papers highlight the importance of data analytics in the healthcare sector. Healthcare data analytics helps in improving patient outcomes, reducing operational costs, and enhancing decision-making processes.

Tools like Microsoft Excel and Power BI are widely used for data cleaning, analysis, and visualization. Excel provides functionalities such as data filtering, pivot tables, and formulas, while Power BI offers advanced visualization capabilities and interactive dashboards.

Previous research also emphasizes the role of dashboards in healthcare management. Dashboards provide a visual representation of key performance indicators, enabling administrators to monitor hospital operations effectively.

In recent years, Artificial Intelligence tools have also been integrated into data analytics processes. AI tools assist in generating insights, automating tasks, and improving efficiency. This project incorporates these concepts by combining data analytics and AI tools to analyze hospital data.

CHAPTER 3: PROPOSED METHODOLOGY

3.1 System Design

The project begins with collecting a hospital dataset from sources such as Kaggle or simulated data. The dataset contains patient-related information including demographic details, diagnosis, treatment outcomes, and hospital department data.

The data is stored in Excel format, which allows easy manipulation and preprocessing. The system is designed to follow a step-by-step approach, starting from data collection to final dashboard creation.

3.2 Data Cleaning and Preprocessing

Data cleaning is one of the most important steps in the project. The raw dataset is examined for errors such as duplicate records, missing values, and inconsistent formats.

Duplicate records are removed to avoid incorrect analysis. Missing values are handled either by filling them with appropriate values or removing incomplete entries. Data formats such as dates are standardized to maintain consistency.

Categorical variables such as gender and diagnosis are also standardized to ensure uniformity across the dataset.

3.3 Data Analysis

After cleaning the dataset, data analysis is performed using Excel and Power BI. Various calculations and formulas are used to derive meaningful insights from the data.

Key performance indicators such as total number of patients, average length of stay, department-wise patient count, and treatment success rate are calculated. These metrics help in understanding hospital operations and patient trends.

3.4 Visualization and Dashboard

An interactive dashboard is created using Power BI to present the analyzed data. The dashboard includes various visual elements such as bar charts, pie charts, line graphs, and KPI cards.

These visualizations help in representing complex data in a simple and understandable manner. Filters and slicers are added to allow users to interact with the dashboard and explore data based on different criteria.

3.5 Requirements

The project requires a laptop or computer with minimum specifications, Microsoft Excel, Power BI Desktop, and a stable internet connection. AI tools such as ChatGPT and Microsoft Copilot are also used to assist in analysis and documentation.

CHAPTER 4: IMPLEMENTATION AND RESULTS

4.1 Data Cleaning Results

The initial dataset contained several issues such as duplicate records, missing values, inconsistent department names, and incorrect data formats. These issues were identified and corrected through data cleaning techniques.

After preprocessing, the dataset became structured and reliable for analysis. This improved the accuracy of the results and ensured meaningful insights.

4.2 Visualization Results

The Power BI dashboard successfully visualizes key healthcare metrics. It provides insights into patient distribution across departments, treatment success rates, and hospital performance trends.

The dashboard allows users to easily understand patterns and make informed decisions. It also helps in identifying areas where improvements are needed.

4.3 AI Integration

Artificial Intelligence tools played a significant role in this project. They were used to generate summaries, explain trends, and assist in report writing.

AI tools improved efficiency and reduced the time required for analysis and documentation. However, the outputs were carefully reviewed to ensure accuracy.

CHAPTER 5: DISCUSSION AND CONCLUSION

The project demonstrates the importance of data analytics in healthcare management. By cleaning and analyzing hospital patient data, valuable insights were obtained that can help improve hospital operations.

The dashboard provides a clear and interactive way to monitor patient data and evaluate performance. It helps hospital administrators make data-driven decisions and improve resource utilization.

Although the project uses a limited dataset, it highlights the potential of data analytics tools in transforming healthcare data. Future improvements can include real-time data integration, advanced analytics, and machine learning models.

In conclusion, this project successfully shows how data cleaning, analysis, and visualization, combined with AI tools, can enhance healthcare decision-making and improve overall efficiency.

REFERENCES

[1] Microsoft Corporation, *Microsoft Excel Documentation*. Available at: https://support.microsoft.com

[2] Microsoft Corporation, *Power BI Documentation*. Available at: https://learn.microsoft.com/power-bi

[3] Edunet Foundation, *AICW Fellowship Program Guidelines*, 2026

[4] Kaggle Inc., *Healthcare Datasets*. Available at: https://www.kaggle.com

[5] OpenAI, *ChatGPT for Data Analysis*. Available at: https://chat.openai.com

[6] Microsoft, *Copilot AI Tool Documentation*. Available at: https://copilot.microsoft.com

[7] Jiawei Han, Micheline Kamber, *Data Mining: Concepts and Techniques*, Morgan Kaufmann

APPENDICES

Appendix A: Dataset Description

The dataset used in this project contains hospital patient records including patient demographics, diagnosis details, treatment outcomes, and hospital department information.

Fields included:

- Patient ID
- Age
- Gender
- Diagnosis
- Admission Date
- Discharge Date
- Department
- Treatment Outcome

Appendix B: Tools Used

- Microsoft Excel (Data Cleaning & Preprocessing)
- Power BI Desktop (Dashboard & Visualization)
- ChatGPT (Documentation & Insights)
- Microsoft Copilot (AI Assistance)

Appendix C: Sample Calculations

Average Length of Stay =

Total Days Stayed ÷ Total Number of Patients

Treatment Success Rate =

(Successful Treatments ÷ Total Treatments) × 100

Appendix D: Dashboard Description

The dashboard includes:

- KPI Cards (Total Patients, Avg Stay)
- Bar Chart (Department-wise Patients)
- Line Graph (Admissions vs Discharges)
- Pie Chart (Treatment Outcomes)
- Filters for Department and Date